

Gyro Logic Type System

1. Primitive Types

World : system state

Observer : observation entity

Query : request for observation

Perception : slice of the world

Goal : objective state

Expectation : observer hypothesis/filter

Conflict : difference between perceptions

Resolution : conflict settlement

History : event log structure

Balance : system stability metrics

Truth : observer-relative evaluation

2. Collection Types

ObserverSet $O = \{o_1, \dots, o_n\}$

PerceptionSet $P = \{p_1, \dots, p_n\}$

ConflictSet C

HistorySeq $H = (h_1, \dots, h_t)$

3. Temporal Types

S_t : World at time t

p_t^o : perception by observer o

g_t : goal at time t

b_t : balance at time t

4. Derived Types

Slice : $\text{World} \times \text{Query} \times \text{Expectation} \rightarrow \text{Perception}$

Eval_G : $\text{Perception} \times \text{Goal} \times \text{Expectation} \rightarrow \text{Truth}$

Detect : $\text{PerceptionSet} \rightarrow \text{ConflictSet}$

Resolve : $\text{Conflict} \times \text{Goal} \times \text{History} \rightarrow \text{Resolution}$

Rebal : $\text{Balance} \rightarrow \text{BalanceDelta}$

5. Delta Types

ObservationDelta

ResolutionDelta

GoalDelta

BalanceDelta

6. Core Operator Types

GenQ : Observer $\times$ World $\rightarrow$ Query

Slice : World $\times$ Query $\times$ Expectation $\rightarrow$ Perception

Eval_G : Perception $\times$ Goal $\times$ Expectation $\rightarrow$ Truth

Compare : Perception $\times$ Perception $\rightarrow$ Distance

Detect : PerceptionSet $\rightarrow$ ConflictSet

Merge : PerceptionSet $\rightarrow$ Perception

UpdateHistory : History $\times$ Event $\rightarrow$ History

Balance : World $\times$ ObserverSet $\times$ History $\rightarrow$ Balance

Rebalance : Balance $\rightarrow$ BalanceDelta